

SCU Analytics Showdown Spring 2026

Group 9 – Instructions

The submission contains five folders. Request you to keep them together at the same level, the Jupyter Notebooks rely on relative paths and may break if the structure is changed.

1. Folder Details

Folder	Contents
Jupyter Notebooks	Four analysis notebooks (run these in order)
Datasets	Raw CSVs (farmer_details, loan_details, planting_survey, buyback_details) plus all computed intermediate files: master_features.csv, risk_tiers.csv, shap_feature_importance.csv, model_comparison_results.csv
JSON Files	Saved model hyperparameters and trained model weights used by the notebooks (best_xgb_params.json, stage1_params.json, stage2_params.json, and associated model files)
Visualizations	15 output charts (01–15) generated by the notebooks, named in the order they appear across the analysis
Python Scripts	Standalone helper scripts used during development. Not required to run the notebooks.

2. Prerequisites

Please ensure the following Python packages are installed before running the notebooks: pandas, numpy, matplotlib, xgboost, sci-kit learn, shap, scipy, seaborn

3. How to Run

Step 1 : Download all five folders and place them together inside a single parent folder

Step 2 : Run the notebooks in order

Step 3 : Charts are saved automatically to the Visualizations/ folder as each notebook runs.

4. Jupyter Notebook Descriptions

00_Data_Preparation.ipynb — *Data Ingestion & Feature Engineering*

Reads the four raw CSVs (farmer_details, loan_details, planting_survey, buyback_details), cleans and merges them, engineers model-ready features (yield, loan aggregates, demographic flags), and saves master_features.csv to the Datasets/ folder. All subsequent notebooks depend on this file : however, master_features.csv is pre-computed and included, so notebooks 01–03 can be run independently.

01_Predictive_Model.ipynb — *Yield Prediction Model*

Trains and benchmarks multiple regression models (XGBoost, Random Forest, Ridge, Lasso) on farmer yield. XGBoost is selected as the final model. Outputs include a model comparison chart, band accuracy analysis (% of farmers predicted within 100/200/500/1000 kg), and a validation residuals plot.

02_Yield_Prediction_Input_Effectiveness.ipynb — *Yield Predictions and Input Effectiveness*

Uses SHAP values from the trained XGBoost model to rank input effectiveness. Covers regional yield analysis, crop risk signals, grade quality by crop, fungicide confounding analysis, fertilizer ROI, and partnership programme impact.

03_Risk_Identification.ipynb — *Farmer Risk Identification*

Builds a two-stage hurdle model to classify farmers into LOW / MEDIUM / HIGH non-seller risk tiers. Covers extension efficiency thresholds, risk tier profiling, tenure analysis, and early-season validation.

5. Important Notes

- Please do not rename or restructure the folders, all paths are relative and depend on the five folders being siblings.
- All intermediate files (master_features.csv, risk_tiers.csv, model params) are pre-computed and included; do not need to re-run earlier notebooks to run later ones.
- Python Scripts/ contains development utilities and does not need to be run.